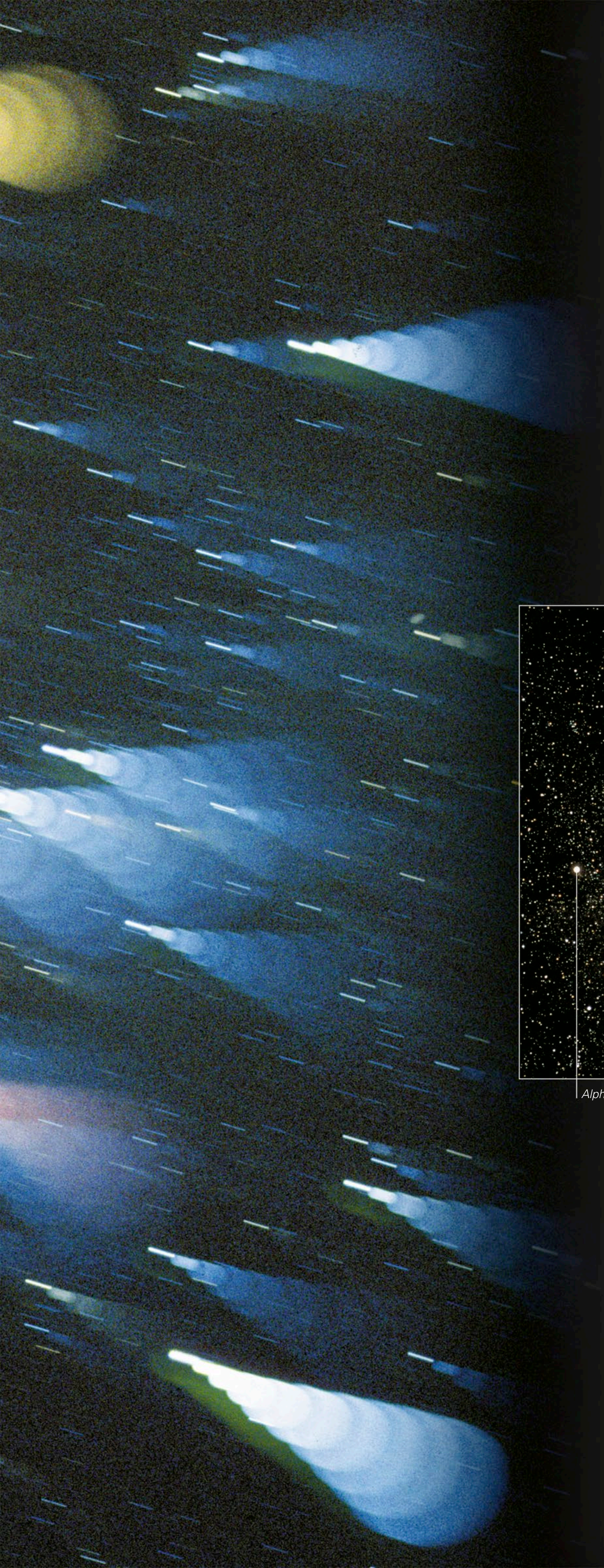
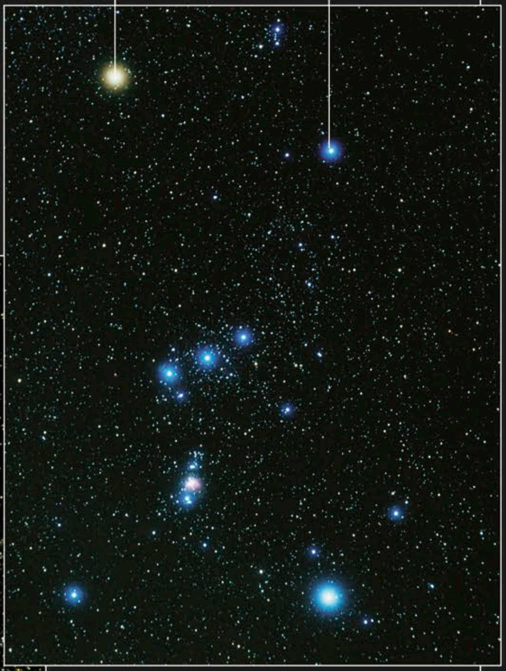


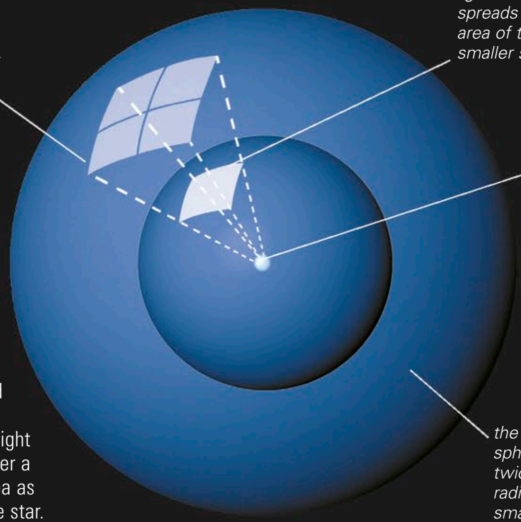
THE BRIGHTNESS OF STARS

A star's brightness in the sky depends on its distance from Earth and on its intrinsic brightness, which is related to its luminosity (the amount of energy it radiates into space per second, see p.233). To compare how stars would look if they were all at the same distance, astronomers use a measure of intrinsic brightness called the absolute magnitude scale. This scale uses high positive numbers to denote dim stars and negative numbers for the brightest ones. A star's brightness as seen from Earth, on the other hand, is described by its apparent magnitude. Again, the smaller the number of a star's apparent magnitude, the brighter the star. Stars with an apparent magnitude of +6 are only just detectable to the naked eye, whereas the apparent magnitude of most of the 50 brightest stars is between +2 and 0. The four brightest (including the brightest star of all, Sirius) have negative apparent magnitudes.



INTRINSICALLY BRIGHT STAR
The stars Betelgeuse and Bellatrix mark the shoulders of Orion. Betelgeuse is noticeably brighter (apparent magnitude 0.45) than Bellatrix (1.64), despite being twice as distant. It is a red, high-luminosity supergiant, whereas Bellatrix is a much less luminous giant.

NEARBY BRIGHT STAR
In the constellation Centaurus, the triple star system Alpha (α) Centauri is a little brighter (apparent magnitude –0.01) than the binary star Hadar, or Beta (β) Centauri (0.61). The reason for Alpha Centauri's brightness is its proximity—it is our closest stellar neighbor. The blue giant stars that make up Hadar are much more luminous than the stars in Alpha Centauri, but they are nearly 100 times farther away.



THE INVERSE SQUARE RULE
The apparent brightness of a star drops in proportion to the square of its distance from the observer—a rule called the inverse square law. This happens because light energy is spread out over a progressively larger area as it travels away from the star.